# Geometric Nonlinear Analysis of a Cantilever Beam with BREP elements

## Objective:

To validate the Geometric Nonlinear Analysis of the shell element with different types of patches in the Kratos IGA application.

## 1. Model Description:

The reference model is a cantilever beam subjected to end shear force, taken from "Popular Benchmark problems for geometric nonlinear analysis of shell" by K.Y. Sze et al. The tip model is given in Fig. 1, and displacement results are shown in Fig. 2. Please refer to the paper for further.

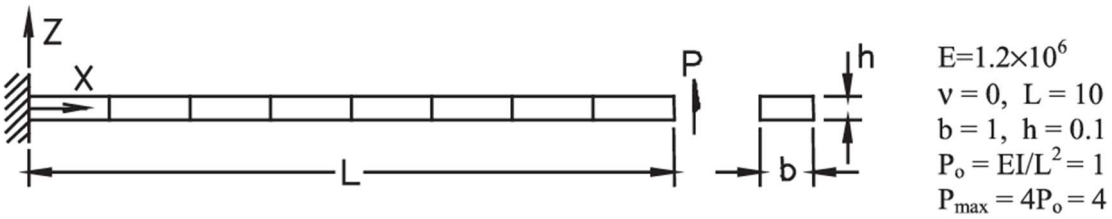



Fig 1: Model

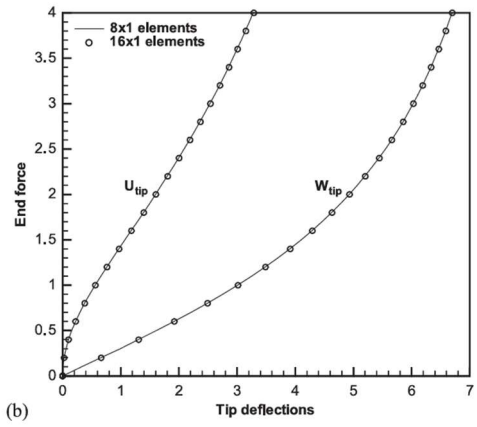



Fig 2: Reference results – Tip Displacement

## 2. Study:

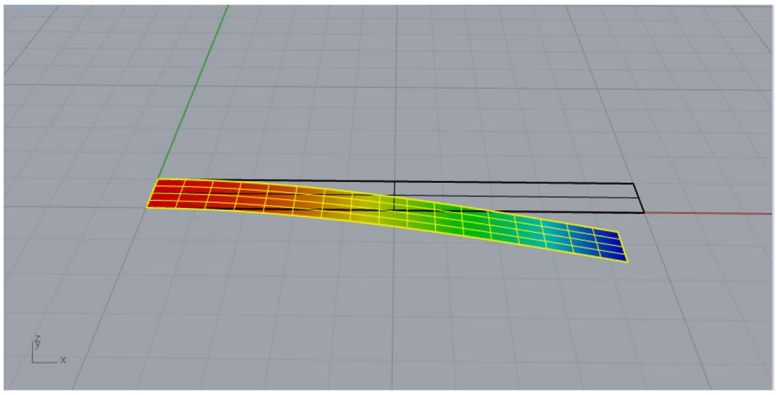
The following types of patches were studied.

1. Single Patch
2. Matching Patches
3. Non-Matching Patches

4. Straight Trimmed Patches
5. Curved Trimmed Patches

## 2.1 Single Patch

In this model, the entire cantilever beam was modelled with a single-patch IGA shell element.

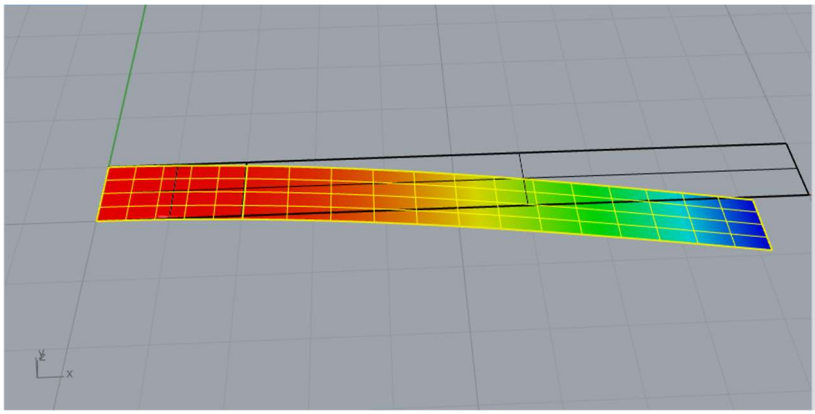


No of control points in X direction = 3
No of control points in Y direction = 16

The peak load (4 units) was reached by iteratively increasing the load by 10 load steps (step size 4/10 = 0.4). In each load step, the solution converged in 3 iterations. Also, if the peak load is selected in single load step, the solver converged within 10 iterations.

## 2.2 Matching Patches

In this model, the entire cantilever beam was modelled with two patches of IGA shell elements. It was ensured that the degree of the curve and the number of control points on the connecting edge match

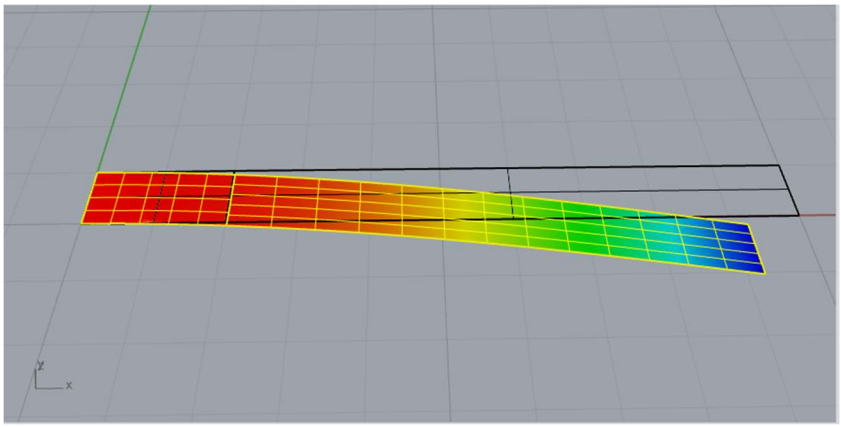


No of control points in X direction = 4,12
No of control points in Y direction = 3,3

The peak load (4 units) was reached by iteratively increasing the load by 10 load steps. In each load step, the solution converged in 6 to 13 iterations. Also, if the peak load is selected in single load step, the solver converged in 17 iterations

### 2.3 Non-Matching patches

In this model, the entire cantilever beam was modelled with two patches of IGA shell elements. But, when compared to the previous model, here the number of control points on the connecting edge is not matched.

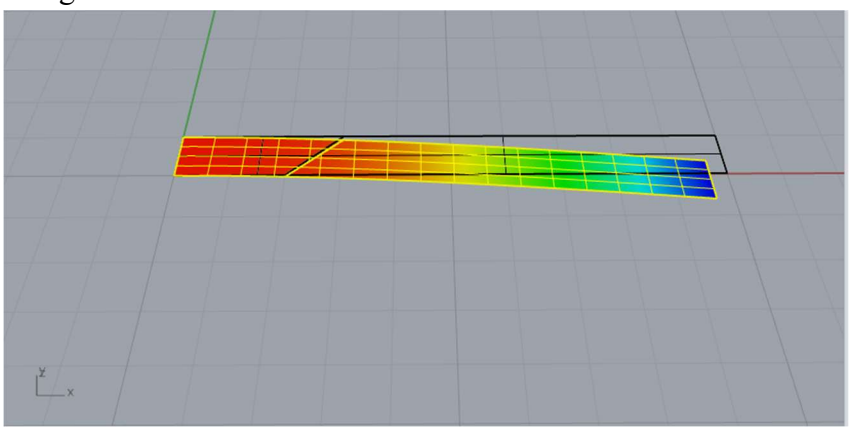


No of control points in X direction = 4,12
No of control points in Y direction = 3,4

The peak load (4 units) was reached by iteratively increasing the load by 10 load steps. In each load step, the solution converged in 6 to 13 iterations. Also, if the peak load is selected in single load step, the solver converged in 17 iterations

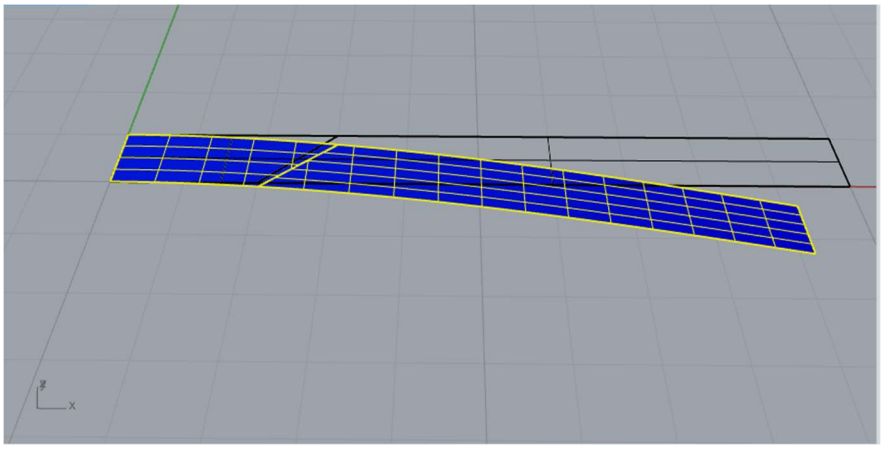
### 2.4 Straight Trimmed Patches

In this model, the entire cantilever beam was modelled with two patches of IGA shell elements. The two IGA elements were trimmed IGA elements. The trimming curve is a straight line



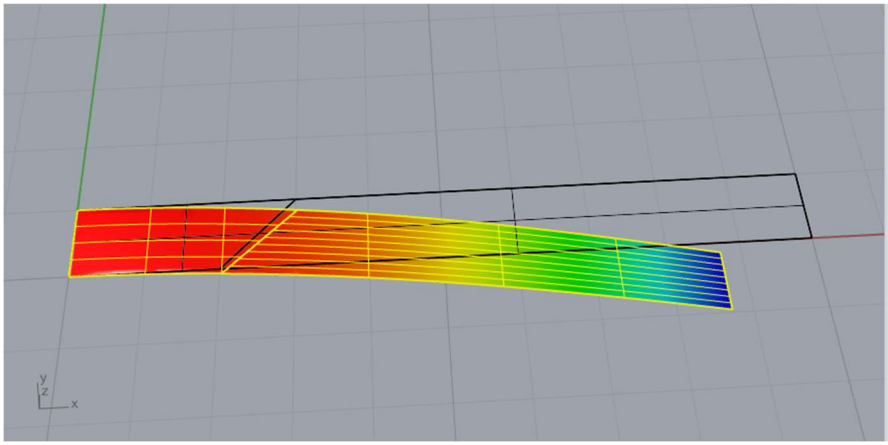
No of control points in X direction = 4,12
No of control points in Y direction = 3,3

For this model, with a load step size of 0.4 and relaxed convergence criteria by 0.1, sometimes the solver converges up to a load value of 0.8. After that, it did not converge. However, with a reduced load step size of 0.2, the solver also converged for the peak load value.



No of control points in X direction = 4,12
No of control points in Y direction = 3,4

If the two patches had non-matching CP, then with the same step size and convergence criteria, the solver only converged up to the load value of 0.4 units. With a reduced step size of 0.04, the convergence was robust up to a load value of 2 units. After that, convergence was not robust; some load steps converged, while others didn't. The step size was further reduced to 0.02. However, even with this load step size, convergence was robust up to a load value of 2.4.



No of control points in X direction = 2,3
No of control points in Y direction = 3,8

If the non-matching patch is used with the above configuration and a reduced step size of 0.02, the solver converges for all steps up to the peak load of 2.4 units.

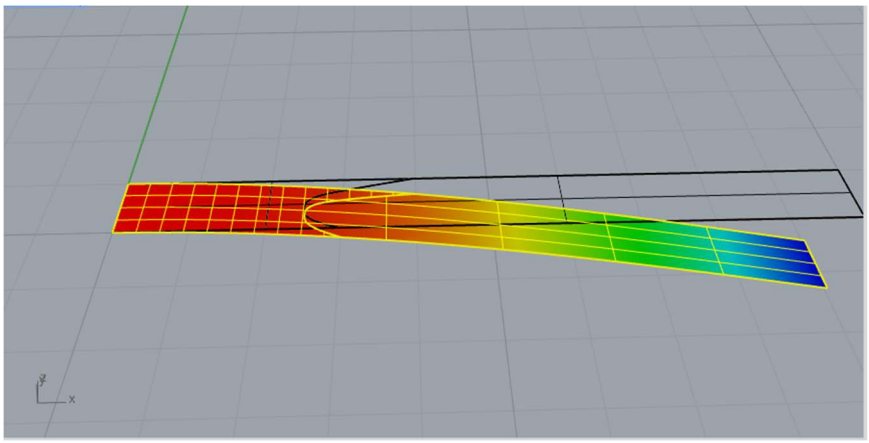
The reason may be that the solver is having difficulty converging after the load value of 2 to 2.5; the load-displacement curve becomes steeper, indicating that the problem becomes stiffer. It might also be observed that if the patch is non-matching, convergence is highly sensitive to the number of CP. If the CP on the X-direction (along the length direction) is reduced, the convergence range of the load value is reduced, which is also intuitive.

## 2.5 Curved Trimmed Patches

In this model, the entire cantilever beam was modelled with two patches of IGA shell elements. The two IGA elements were trimmed IGA elements. The trimming curve is a curve.

No of control points in X direction = 4, 12
No of control points in Y direction = 3, 3

Unable to achieve convergence after a load value of 0.4, even with the load step value. It is a counterintuitive observation.



No of control points in X direction = 12, 4
No of control points in Y direction = 3, 3

With a load step of 0.02, the convergence was achieved in load steps up to a load value of 2.4 to 2.8 units. After that, convergence was non-robust.

No of control points in X direction = 4, 12

No of control points in Y direction = 3, 3

Unable to achieve convergence even for the load value of 0.4, even with the load step value. It is a counterintuitive observation.

No of control points in X direction = 12, 4
No of control points in Y direction = 4, 3

With a load step of 0.02, the convergence was achieved in load steps up to a load value of 2.4 to 2.8 units. After that, convergence was non-robust.